

# HAT-P-36b

R-0698

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-36\_260331 · created 2026-07-27T21:25:39+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2461130.7389 +/- 0.0067 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.063 +/- 0.041                 |
| Transit depth (Rp/Rs)^2                           | 0.39 +/- 0.52 [%]               |
| Orbital Inclination (inc)                         | 80.25 +/- 2.83                  |
| Airmass coefficient 1 (a1)                        | 0.864 +/- 0.046                 |
| Airmass coefficient 2 (a2)                        | 0.13 +/- 0.18                   |
| Scatter in the residuals of the lightcurve fit is | 5.25 %                          |
| Best Comparison Star                              | #3 - [607, 234]                 |
| Optimal Aperture                                  | 2.54                            |
| Optimal Annulus                                   | 8.04                            |
| Transit Duration (day)                            | 0.06 +/- 0.022                  |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                          |
|---------------------------|------------------------------------------|
| Rp/R■ fit vs published    | 0.063 ± 0.041 vs 0.126 (deviation 1.54σ) |
| Duration fit vs published | 0.06 d vs 0.0925 d (deviation 1.48σ)     |
| Mid-time O–C              | -11.9 min                                |
| Depth SNR                 | 1.5                                      |
| Residual scatter          | 5.25 %                                   |

### Light curve

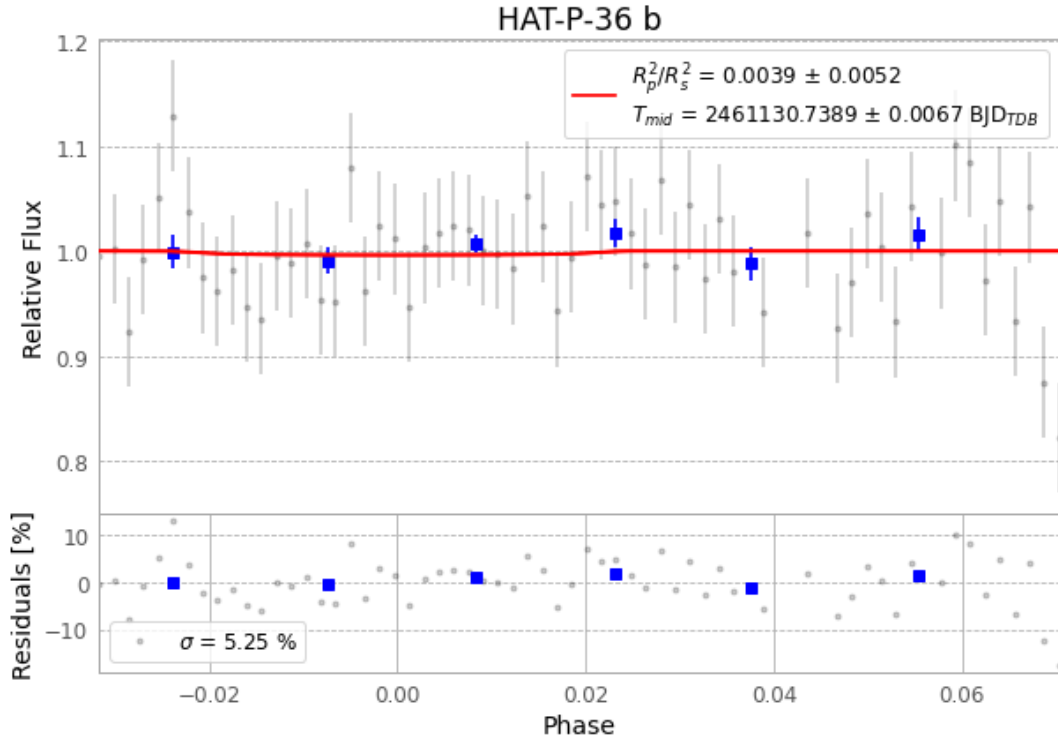


Figure 1 — FinalLightCurve\_HAT-P-36 b\_2026-03-30.png (SHA-256 99d33761b2077a5e...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [276, 256], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 6cb7a33f6bfc46aa...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 62f7813

## Data provenance

68 FITS frames (45 MB), HATP-36260331043616.FITS → HATP-36260331075715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260331.log (7b6d59a2bbca...), FinalLightCurve\_HAT-P-36 b\_2026-03-30.png (99d33761b207...), FinalLightCurve\_HAT-P-36 b\_2026-03-30.csv (17522e451016...)

## Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:25Z.